Leakage-aware drug discovery workflow for PKM2 and MAPK1

The study combines dataset auditing, scaffold-aware benchmarking, calibrated ranking, ADMET triage, docking and pose-level contact checks.

Key principle

No single score is treated as a discovery claim.

1. Target extraction and curation

LIT-PCBA subsets for PKM2 and MAPK1
Canonical SMILES audit
Duplicates and label conflicts removed

2. EDA and scaffold mapping

Descriptor distributions
Class imbalance inspection
Bemis-Murcko scaffold coverage

3. Benchmark, modelling and scoring

Nearest-active Tanimoto baseline
Random/scaffold split evaluation
AP, EF1%, BEDROC20 and repeated seeds

Core workflow

4. Calibration and uncertainty

Isotonic probability mapping
ECE10, Brier score, log-loss
Ensemble disagreement as ranking support

5. ADMET-aware shortlist and structure-aware triage

Drug-likeness filters
ADMET-AI support score
AutoDock Vina docking and residue-contact checks

Downstream triage

Target-specific interpretation

PKM2 branch

Stronger scaffold-level ML gains
Coherent support across calibration, ADMET and docking

MAPK1 branch

Biologically important contrast target
Similarity remains competitive and downstream triage becomes more decisive